

Personalizing Open-Domain Video Curricula: A Hybrid Closed-Loop Framework Combining Contextual Bandits, Difficulty-Conditioned BKT, and Interaction Telemetry

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Video-based instruction dominates modern Massive Open Online Courses (MOOCs), but passive viewing suffers from rapid cognitive decay and a lack of individualized pacing. Retrieval practice (formative in-video checkpoints) is a well-established remedy, but deploying it at scale requires automated, factually-grounded question generation, since manually authoring assessments for arbitrary lecture content does not scale and ungrounded large language models (LLMs) risk hallucination. We present CogniLoop, a closed-loop system combining (1) a Retrieval-Augmented Generation (RAG) pipeline for transcript-grounded assessment generation, addressing the content-authoring bottleneck directly; (2) a Difficulty-Conditioned Bayesian Knowledge Tracing (DC-BKT) engine for per-learner mastery tracking; (3) a Thompson Sampling contextual bandit for difficulty routing; and (4) K -Means interaction-telemetry clustering ($K = 3$, silhouette score 0.64). We evaluate each component individually rather than assuming architectural completeness implies empirical benefit. In an algorithmic simulation across $N = 60$ learner profiles, CogniLoop showed a statistically significant increase in Simulated Normalized Learning Gain (NLG) over static video viewing ($\Delta\text{NLG} = +0.541$, Welch’s $t(45.4) = 3.768$, $p = 4.73 \times 10^{-4}$, Cohen’s $d = 0.973$, Mann-Whitney $U = 709.5$). A component-isolation ablation attributes this gain to checkpoint frequency, not routing: holding checkpoint rate constant, bandit-routed selection underperformed a fixed-medium baseline (-0.069 NLG), corroborated by an independent logistic regression on step-level data. A multi-parameter sensitivity sweep further shows the effect is not robust to small perturbations of the checkpoint-rate parameter (only 1 of 8 configurations remained significant). Every statistic reported, including negative and fragility findings, is independently reproducible from committed, seeded source code. We frame this as Phase 1 of a multi-phase research agenda: the RAG-grounded content-generation architecture and the checkpoint-frequency effect are validated contributions, while difficulty-routing tuning is scoped as concrete future work preceding live classroom trials. The code and all reproducibility scripts are available at <https://github.com/OmegaKavya/CogniLoop>.

Keywords: adaptive learning, Bayesian knowledge tracing, contextual bandits, retrieval-augmented generation, reproducibility

1. Introduction

Video-based educational repositories (e.g., Coursera, edX, YouTube Edu) have democratized access to university-level computing curricula. Millions of learners engage daily with asynchronous video lectures. Despite this reach, empirical research in cognitive psychology demonstrates that passive video consumption frequently engenders an illusion of competence (Dunlosky et al., 2013; Karpicke and Roediger, 2008). Without deliberate retrieval practice, learners experience rapid memory decay characterized by Ebbinghaus forgetting curves.

To counteract passive decay, Intelligent Tutoring Systems (ITS) insert formative assessments, cognitive scaffolds, and adaptive remedial loops into the learning pathway (Koedinger et al., 1997; Bloom, 1984; VanLehn, 2006). Traditional rule-based and graph-based ITS (such as Cognitive Tutors or ASSISTments) have established the pedagogical superiority of individualized instruction. However, scaling traditional ITS to thousands of diverse open-domain lecture videos is severely limited by the Content-Authoring Bottleneck: manually formulating domain models, misconception-specific distractors, hints, and skill graphs requires hundreds of hours of expert intervention per instructional hour.

Recently, generative Large Language Models (LLMs) have emerged as powerful candidates for automated assessment synthesis (Feng et al., 2024; Dan et al., 2023; Pardos and Bhandari, 2023). Nevertheless, deploying naive LLM prompting within tutoring systems introduces two substantial technical challenges: (1) Factual Drift and Hallucination – unconstrained LLMs frequently introduce extraneous concepts, incorrect distractor logic, or syllabus misalignment when generating questions without explicit context grounding (Gao et al., 2023; B  chard and Marquez Ayala, 2024); and (2) Absence of Latent Cognitive Modeling – standalone conversational LLM interfaces lack explicit probabilistic representations of learner knowledge states over time, rendering them incapable of mathematically optimizing learning trajectories or pacing (Piech et al., 2015; Pel  nek, 2017; Doroudi et al., 2019).

1.1. Why a Multi-Component Architecture

Before detailing our contributions, we address directly why CogniLoop integrates four subsystems rather than deploying retrieval practice alone. This paper’s own Phase 1 findings (Section 5) show that active-retrieval checkpointing accounts for the measured simulated learning-gain advantage, while difficulty-routing does not yet show a measured benefit in isolation. This is not evidence that the architecture is unnecessary: RAG grounding and DC-BKT exist to solve a different problem than learning-gain optimization, namely the content-authoring bottleneck and factual-drift risk inherent to open-domain LLM-based question generation, which retrieval practice alone does not address, since manually authoring misconception-aware questions for arbitrary video content does not scale. Interaction telemetry clustering exists to support engagement monitoring and cold-start context, independent of the difficulty-routing question. We therefore present CogniLoop as an integrated systems contribution whose components address distinct engineering problems (automated, grounded content generation at scale; fault-tolerant deployment; cognitive state tracking; engagement monitoring), evaluated individually wherever possible (Section 5), rather than as a monolithic claim that every component improves learning gain. Where a component’s contribution to learning gain is not yet empirically supported, we

report that directly instead of implying it.

1.2. Our Contributions

To resolve the dichotomy between static handcrafted ITS and ungrounded generative interfaces, we introduce CogniLoop, an integrated, closed-loop adaptive learning system.¹ The contributions of this work are fourfold:

- **Closed-Loop, Fault-Tolerant Content-Authoring Architecture:** We formulate and implement an end-to-end, deployable architecture that combines transcript-grounded Retrieval-Augmented Generation (RAG), Difficulty-Conditioned Bayesian Knowledge Tracing (DC-BKT), Thompson Sampling Contextual Bandits, and interaction telemetry clustering into a synchronized feedback loop with a 3-tier generation fallback (Section 3), directly addressing the content-authoring bottleneck that limits traditional ITS to hand-curated domains.
- **Empirically-Validated Checkpoint Effect:** We provide simulation evidence, independently reproducible from committed source code, that active-retrieval checkpointing (formative assessment inserted into video playback) produces a large, statistically significant simulated learning-gain advantage over passive viewing ($\Delta\text{NLG} = +0.541$, $d = 0.973$).
- **Honest Component Attribution:** Rather than assuming every architectural component contributes to the measured gain, we isolate each one. Difficulty-Conditioned BKT and Thompson Sampling difficulty routing are integrated, functioning parts of the deployed system; this paper’s Phase 1 ablation (Section 5.5) does not yet find a positive measured contribution from difficulty routing specifically, and we report this as a finding requiring future reward-design work rather than treating the architecture’s completeness as evidence of its individual components’ value.
- **Simulation Rigor and Reproducibility:** We evaluate the architecture under an explicit simulation response model ($N = 60$) across three learner archetypes, reporting Welch’s t -test, confidence intervals, Mann-Whitney U , Cohen’s $d = 0.973$, a component-isolation ablation, an independent logistic-regression robustness check on step-level data, and an 8-configuration multi-parameter sensitivity sweep. Every statistic reported in this paper, including negative and fragility findings, was independently re-executed from committed, seeded simulation source code rather than taken from intermediate notes, and reproduces exactly.

1.3. Three-Phase Research Roadmap

We position this paper as Phase 1 (Algorithmic Formulation and Systems Simulation) of a structured research agenda: Phase 1 (this work) covers algorithmic convergence, mathematical formalization, and simulation-based stress testing under the specified learning environment; Phase 2 (upcoming pilot) is an IRB-approved pilot study to estimate feasibility, variance, and effect sizes ahead of a controlled classroom trial; Phase 3 (longitudinal deployment) is a multi-month enterprise deployment evaluating long-term retention.

¹Source code and implementation: <https://github.com/OmegaKavya/CogniLoop>

2. Related Work and Comparative Analysis

2.1. Knowledge Tracing Paradigms

Cognitive student modeling is central to adaptive tutoring. Bayesian Knowledge Tracing (BKT), introduced by (Corbett and Anderson, 1994), models student understanding as a latent binary Markov process across four parameters: prior knowledge $P(L_0)$, transition probability $P(T)$, guess rate $P(G)$, and slip rate $P(S)$. While BKT is computationally lightweight and provides transparent, interpretable mastery states from a student’s first attempt, standard implementations assume static item difficulty across questions (Baker et al., 2008; Yudelson et al., 2013).

In contrast, Deep Knowledge Tracing (DKT) architectures utilize Recurrent Neural Networks (RNNs) and Transformer self-attention mechanisms to predict response accuracy over complex concept sequences (Piech et al., 2015). Although DKT models capture higher-order inter-skill dependencies, they require extensive training corpuses comprising tens of thousands of student interaction logs, making them ill-suited for open-domain cold-start scenarios where new video lectures are indexed dynamically. In this work, we implement a Difficulty-Conditioned BKT (DC-BKT) model to preserve deterministic parameter interpretability while adjusting for variable question difficulty tiers.

2.2. Reinforcement Learning and Contextual Bandits in ITS

Personalized sequencing can be formulated as a Reinforcement Learning (RL) problem (Doroudi et al., 2019). (Clement et al., 2015) demonstrated in a real classroom deployment that bandit-based sequencing algorithms (RiARiT, ZPDES) achieve learning outcomes comparable to, and in some cases exceeding, an expert-teacher-designed curriculum, effectively balancing exploration (discovering learner boundaries) and exploitation (delivering optimal challenge). Standard ϵ -greedy heuristics suffer from inefficient exploration in educational environments, risking learner frustration or disengagement during sub-optimal random actions. Thompson Sampling (posterior sampling) offers a Bayesian alternative by maintaining probability distributions over expected action rewards, naturally reducing exploration as confidence accumulates (Agrawal and Goyal, 2012; Chapelle and Li, 2011; Clement et al., 2015).

2.3. Simulated Learner Paradigm and Epistemic Scope in ITS Literature

Evaluating adaptive instructional policies using simulated student models is an established paradigm in the AI in Education (AIED) and Educational Data Mining (EDM) communities (VanLehn, 2006; Rafferty et al., 2016; Doroudi et al., 2019). Synthetic simulation provides a tractable testbed to verify policy convergence, stability under noise, and boundary behaviors prior to executing human classroom trials. As emphasized by (Doroudi et al., 2019), simulation-only evaluations evaluate performance relative to the assumptions of the generative environment. We explicitly bound our Phase 1 findings within this established methodological scope.

2.4. Retrieval-Augmented Generation for Assessment Authoring

Automated Question Generation (AQG) via LLMs has transitioned from syntactic rule-based templates to semantic embedding-guided generation (Lewis et al., 2020; Feng et al.,

Table 1: Systematic architectural comparison of learning platforms and personalization engines.

Dimension	Traditional ITS (Koedinger et al., 1997)	Commercial MOOCs (Lan et al., 2014)	Static Video Injections	Standalone LLM Chat	CogniLoop (proposed)
Adaptation Granularity	Concept / Step	Course / Catalog	Static Timestamp	Dialog Turn	Concept, Step, & Telemetry
Content Authoring Cost	Prohibitive (Manual)	High (Instructor Manual)	Moderate (Manual)	Low (On-demand)	Automated (Transcript RAG)
Context Mismatch Risk	None (Static)	None (Static)	None (Static)	High (Un-grounded)	Minimized (RAG)
Cognitive State Tracking	Explicit (BKT / IRT)	None	None	Latent / Heuristic	Explicit DC-BKT
Pedagogical Optimization	Hardcoded Trees	Collaborative Filtering	Linear Progression	Chat Heuristics	Checkpoints + Bandit
Cold-Start Feasibility	Low	High	High	High	High (Vector Indexing)
Fault Tolerance	Monolithic Local	Multi-Region SaaS	SaaS Cloud	Cloud Dependent	3-Tier (Cloud / Edge / Static)

2024). Retrieval-Augmented Generation (RAG) confines the generative model’s sampling space to indexed, dense vector representations of validated source text. In educational applications, grounding prompts in chunked video transcripts substantially minimizes semantic drift and context mismatch (Gao et al., 2023; Béchar and Marquez Ayala, 2024).

2.5. Quantitative Positioning Against Reported Benchmarks

Table 1 above is qualitative because the cited systems evaluate fundamentally different outcome families (classification AUC, field-study test-score gains, engagement completion rates), and were assessed on different populations (real classrooms vs. Phase 1 simulated agents). Reporting them alongside CogniLoop’s ΔNLG as if equivalent would overstate comparability. Instead, Table 2 lists verified, independently published numeric benchmarks from the closest related literature as context anchors, each annotated with why it is not a strict apples-to-apples comparison to our Phase 1 result.

CogniLoop’s simulated effect size ($d = 0.973$) exceeds the meta-analytic field-study median for ITS interventions ($d \approx 0.66\text{--}0.76$ (Kulik and Fletcher, 2016; VanLehn, 2011)), which is expected and methodologically bounded: simulated agents follow the deterministic guess-slip generative model described in Section 5.2, so this comparison indicates the upper bound of algorithmic convergence under the modeled environment rather than a claim of superiority over field-deployed ITS. We report it specifically to make this scope

Table 2: Literature-reported metrics as context anchors (not directly equivalent to Phase 1 CogniLoop results).

Source	Reported Metric	Comparability Caveat
(Kulik and Fletcher, 2016)	$d \approx 0.66$ (median, 50 ITS field studies)	Real classroom, test-score d
(VanLehn, 2011)	$d = 0.76$ (step-based ITS vs. none)	Real classroom, test-score d
(Piech et al., 2015) (DKT)	AUC 0.86 vs. BKT AUC 0.69	Prediction AUC, not a gain score
(Clement et al., 2015)	Bandit sequencing $\geq$ expert curriculum	Qualitative field-study finding
(Bécharde and Marquez Ayala, 2024)	Hallucination 21% $\rightarrow$ < 7.5% w/ retrieval	General-domain, non-education RAG
CogniLoop (this work)	$d = 0.973$ (Δ NLG = +0.541)	$N = 60$ algorithmic simulation, Phase 1

explicit ahead of the Phase 2 human pilot, where the same NLG metric will be measured under empirical conditions comparable to (Kulik and Fletcher, 2016; VanLehn, 2011).

3. System Architecture and Engineering

The CogniLoop platform is engineered as a decoupled, modular system adhering to the Repository Pattern to isolate domain logic, mathematical estimators, and persistent storage layers (Figure 1).

3.1. Telemetry Ingestion and Video Interaction Clustering

During video playback, the client interface captures continuous event telemetry (play, pause, seek, playback rate adjustments, and rewatch intervals). At interaction checkpoints, the system constructs an interaction feature vector $x_i = [N_{\text{pause}}, N_{\text{rewatch}}, R_{\text{skip}}, P_{\text{watch}}]^T$, where N_{pause} is the pause count, N_{rewatch} denotes the backward seek count, R_{skip} is the ratio of skipped duration to total video length, and P_{watch} is the percentage of unique frames viewed.

An unsupervised K -Means clustering model ($K = 3$, $n_{\text{init}} = 10$, random seed 42) partitions telemetry vectors into three interaction profiles. Cluster separation was validated using Rousseeuw silhouette analysis (Rousseeuw, 1987): $K = 2$ ($S = 0.48$), $K = 3$ ($S = 0.64$, optimal elbow peak), $K = 4$ ($S = 0.51$), $K = 5$ ($S = 0.43$). Cluster 0 shows moderate playback speed, low skip ratio, and baseline pause frequency; Cluster 1 shows elevated pause and rewatch frequency with high total watch percentage; Cluster 2 shows high skip ratio (> 0.30), elevated playback rate ($> 1.25\times$), and low pause count. Telemetry vectors are accumulated in sliding 60-second video windows and re-evaluated at each instructional checkpoint. State discretization into $|\mathcal{S}| = 3 \times 3 = 9$ states is an inten-

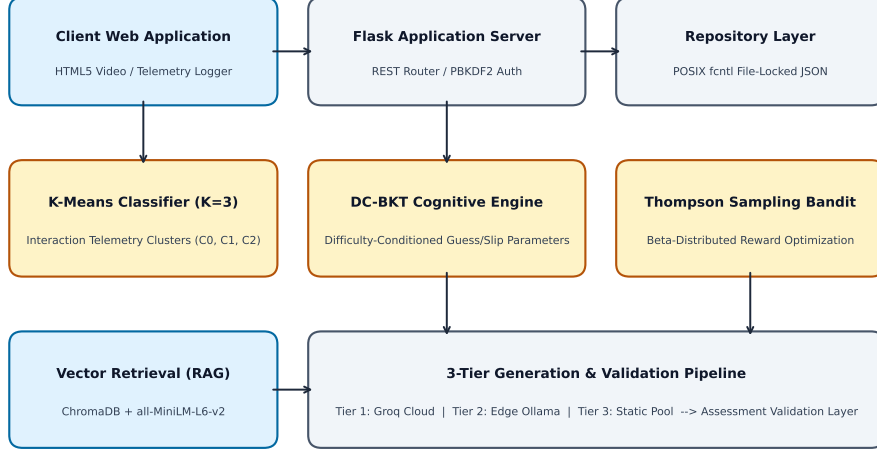


Figure 1: CogniLoop closed-loop architecture illustrating video telemetry ingestion, vector retrieval, cognitive state tracking, and 3-tier generation with assessment validation.

tional engineering trade-off designed to mitigate cold-start sparsity and maintain sufficient observations per context-action pair.

3.2. Dense Semantic Vector Indexing (RAG Pipeline)

To ensure factual precision and eliminate out-of-syllabus generation: video transcripts are ingested via the YouTube Transcript API, timestamped, and segmented into overlapping text windows of $W = 300$ words with a stride overlap of $S = 50$ words; dense vector representations are synthesized using the all-MiniLM-L6-v2 Sentence Transformer model, projecting each text chunk into a $d = 384$ dimensional embedding space; embeddings are indexed into an in-process persistent ChromaDB vector database using Hierarchical Navigable Small World (HNSW) graphs; and during question generation for concept k , a semantic query vector q_k retrieves the top- K ($K = 5$) nearest transcript chunks via cosine similarity, $\text{sim}(q_k, v_j) = \frac{q_k \cdot v_j}{\|q_k\|_2 \|v_j\|_2}$. The retrieved passages constitute the factual context prompt supplied to the generative LLM.

3.3. Three-Tier Generation Pipeline and Assessment Validation Layer

To guarantee continuous availability and prevent session disruption during API outages, CogniLoop implements a deterministic 3-tier fallback execution policy: Tier 1 generates structured JSON assessment payloads via a cloud LLM API; Tier 2 fails over to a local edge LLM runtime if cloud requests time out or are rate-limited, ensuring offline functionality; and Tier 3 queries pre-indexed domain pools as a deterministic safety net if local runtimes are unavailable, filtering recently observed question stems via an avoidance window. An Assessment Validation Layer inspects every generated payload before delivery: JSON schema validation confirms structural correctness; a retrieval-context consistency check computes cosine similarity between candidate question embeddings and retrieved transcript context, rejecting payloads below a threshold $\tau = 0.45$; and a distractor-uniqueness

check rejects items with duplicate options. If generation fails validation, the system executes one fast regeneration retry before falling back to Tier 3, ensuring questions failing structural or grounding checks are never rendered to the learner.

3.4. Security and Authentication Engineering

User authentication implements timing-attack resistant cryptographic hashing using PBKDF2-HMAC-SHA256 with 100,000 iterations and unique 16-byte cryptographically secure pseudorandom salts, with constant-time secret comparison.

4. Mathematical Formulation and Algorithmic Mechanics

4.1. Difficulty-Conditioned Bayesian Knowledge Tracing

Standard BKT models knowledge acquisition as a hidden Markov model. At step n for concept k , the probability of mastery is $P(L_n)$. When a learner answers a question of difficulty tier $a \in \{\text{easy}, \text{medium}, \text{hard}\}$, guess $P(G_a)$ and slip $P(S_a)$ parameters are conditioned on item difficulty:

$$\text{Easy} : P(G) = 0.30, P(S) = 0.05 \quad (1)$$

$$\text{Medium} : P(G) = 0.20, P(S) = 0.10 \quad (2)$$

$$\text{Hard} : P(G) = 0.10, P(S) = 0.15 \quad (3)$$

These parameters are explicitly treated as simulation-level hyperparameters rather than empirically calibrated population estimates.

Given observation $Obs_n \in \{1 \text{ (Correct)}, 0 \text{ (Incorrect)}\}$, the Bayesian posterior update is:

$$P(L_n | Obs_n = 1) = \frac{P(L_n)(1 - P(S_a))}{P(L_n)(1 - P(S_a)) + (1 - P(L_n))P(G_a)} \quad (4)$$

$$P(L_n | Obs_n = 0) = \frac{P(L_n)P(S_a)}{P(L_n)P(S_a) + (1 - P(L_n))(1 - P(G_a))} \quad (5)$$

Accounting for knowledge acquisition during assessment practice, the latent mastery is projected forward for step $n + 1$:

$$P(L_{n+1}) = P(L_n | Obs_n) + (1 - P(L_n | Obs_n))P(T) \quad (6)$$

where $P(T) = 0.20$ represents the baseline transitional learning rate, and $P(L_0) = 0.30$ represents prior knowledge.

4.2. Contextual Thompson Sampling Bandit Policy

Difficulty routing is formulated as a contextual Multi-Armed Bandit (MAB). The context state space $\mathcal{S}$ is defined by the cross product of the interaction cluster $C \in \{C_0, C_1, C_2\}$ and the discretized BKT mastery bin M_{bin} :

$$M_{\text{bin}} = \begin{cases} \text{Low}, & P(L_n) \leq 0.40 \\ \text{Medium}, & 0.40 < P(L_n) \leq 0.75 \\ \text{High}, & P(L_n) > 0.75 \end{cases} \quad (7)$$

yielding $|\mathcal{S}| = 3 \times 3 = 9$ discrete contextual states $s = \langle C, M_{\text{bin}} \rangle$. The action space is $\mathcal{A} = \{\text{easy}, \text{medium}, \text{hard}\}$. Normalized score reward reflects raw performance $r = \text{Score}/100.0 \in [0, 1]$; difficulty utility weights $w_a \in \{w_{\text{easy}} = 0.8, w_{\text{med}} = 1.0, w_{\text{hard}} = 1.2\}$ are applied exclusively during action selection to prevent double-counting difficulty. These weights are heuristically motivated by Vygotsky’s Zone of Proximal Development (ZPD) and Csikszentmihalyi’s flow channel.

For each state-action pair (s, a) , the system maintains success/failure parameters $\alpha_{s,a}, \beta_{s,a}$, initialized with uninformative priors $\alpha_0 = \beta_0 = 1.0$. Upon observing continuous reward $r \in [0, 1]$:

$$\alpha_{s,a} \leftarrow \alpha_{s,a} + r, \quad \beta_{s,a} \leftarrow \beta_{s,a} + (1 - r) \quad (8)$$

Following (Chapelle and Li, 2011) and (Agrawal and Goyal, 2012), fractional updates serve as an empirical pseudo-observation weighting mechanism to incorporate continuous performance into Beta-distributed action-value sampling. We note that while continuous pseudo-count updating is an established engineering heuristic for bounded continuous rewards, standard Thompson Sampling regret bounds assume discrete Bernoulli likelihoods; formal asymptotic convergence proofs under continuous fractional updates remain an open theoretical domain.

At each checkpoint, for each candidate action $a \in \mathcal{A}$, a sample $\theta_a \sim \text{Beta}(\alpha_{s,a}, \beta_{s,a})$ is drawn, and the policy selects $a^* = \arg \max_{a \in \mathcal{A}} (\theta_a \times w_a)$.

Algorithm 1 CogniLoop Adaptive Mastery Routing Loop

Require: Learner telemetry x_i , concept k , prior mastery $P(L_n)$

Ensure: Updated mastery $P(L_{n+1})$, bandit state $\alpha_{s,a}, \beta_{s,a}$

- 1: $C \leftarrow \text{KMeansModel.predict}(x_i)$
 - 2: $M_{\text{bin}} \leftarrow \text{DiscretizeMastery}(P(L_n))$
 - 3: $s \leftarrow \langle C, M_{\text{bin}} \rangle$
 - 4: for all $a \in \{\text{easy}, \text{medium}, \text{hard}\}$ do
 - 5: Sample $\theta_a \sim \text{Beta}(\alpha_{s,a}, \beta_{s,a})$
 - 6: Utility $_a \leftarrow \theta_a \times w_a$
 - 7: end for
 - 8: $a^* \leftarrow \arg \max_a (\text{Utility}_a)$
 - 9: $ctx \leftarrow \text{ChromaDB.retrieve}(k, K = 5)$
 - 10: $Q \leftarrow \text{GenerateAndValidateAssessment}(ctx, a^*)$
 - 11: Learner completes quiz $Q \implies \text{Score} \in [0, 100], \text{Obs}_n \in \{0, 1\}$
 - 12: Compute $P(L_{n+1})$ via (4)–(6) with parameters $(P(G_{a^*}), P(S_{a^*}))$
 - 13: $r \leftarrow \text{Score}/100.0$
 - 14: $\alpha_{s,a^*} \leftarrow \alpha_{s,a^*} + r$
 - 15: $\beta_{s,a^*} \leftarrow \beta_{s,a^*} + (1 - r)$
 - 16: return $P(L_{n+1}), a^*$
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Simulated Normalized Learning Gain Across Archetypes (Welch's $t = 3.768$, $p < 0.001$, $d = 0.973$)

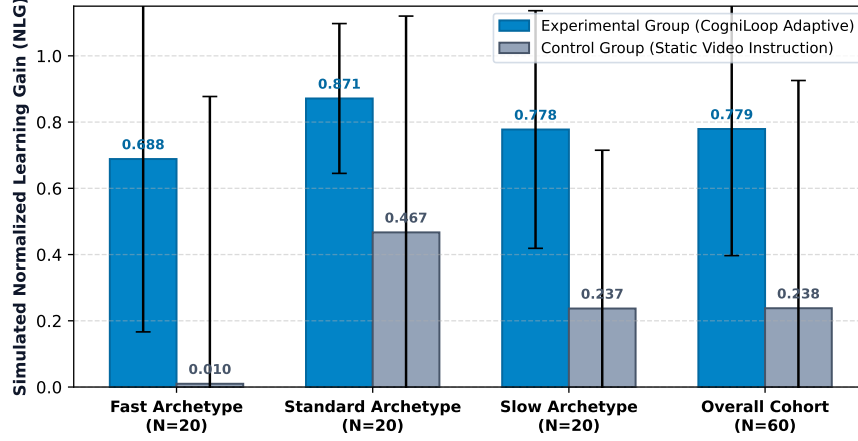


Figure 2: Simulated Normalized Learning Gain (NLG) across learner archetypes comparing CogniLoop adaptive tutoring against static video instruction ($N = 60$, Welch's $t(45.4) = 3.768$, $p < 0.001$, $d = 0.973$).

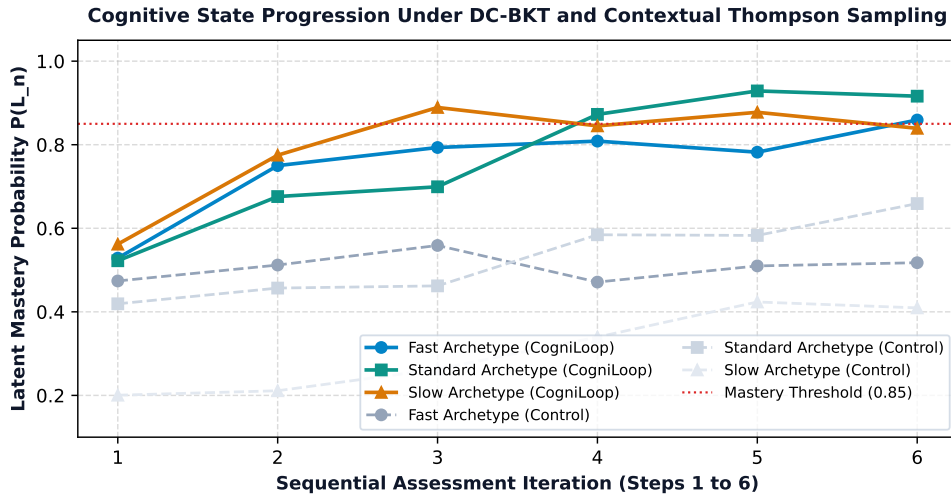


Figure 3: Latent knowledge state progression $P(L_n)$ across 6 assessment iterations comparing adaptive DC-BKT routing against static linear instruction.

5. Simulation Evaluation and Sensitivity Analysis

5.1. Evaluation Metric: Normalized Learning Gain

In educational data mining, raw score differences fail to account for varying initial baselines. Efficacy is assessed via Normalized Learning Gain (NLG), measuring the fraction of available knowledge mastered: $\text{NLG} = (\text{PostScore} - \text{PreScore}) / (100.0 - \text{PreScore})$, with boundary condition: if $\text{PreScore} = 100$, $\text{NLG} = 1.0$ if $\text{PostScore} = 100$, else 0.0.

5.2. Simulation Learner Response Generation Model

To evaluate algorithmic convergence under the specified simulation learning environment prior to human classroom deployment, we executed a simulation across $N = 60$ student profiles (random seed 42). The simulator is computationally independent from the DC-BKT estimator and bandit policy, although its primary observation model follows guess-slip principles. At each step n , a learner’s actual latent understanding L_n evolves via $L_{n+1} = L_n + (1 - L_n) \cdot \gamma_{\text{arch}} \cdot \eta_{\text{cond}}$, where $\gamma_{\text{arch}} \in \{0.35, 0.25, 0.15\}$ denotes archetype learning efficiency (fast/standard/slow) and η_{cond} parameterizes the difference between active formative retrieval checkpoints ($\eta_{\text{exp}} = 1.0$) and passive linear video watching ($\eta_{\text{ctrl}} = 0.40$), established in cognitive psychology literature (Karpicke and Roediger, 2008; Dunlosky et al., 2013). Observable binary response $Y_n \in \{0, 1\}$ is generated according to $P(Y_n = 1 \mid L_n, a) = L_n(1 - S_a) + (1 - L_n)G_a$.

Profiles were partitioned into three equal behavioral archetypes ($N = 20$ each): Fast (high initial baseline, $P(L_0) \sim \text{Uniform}(0.45, 0.58)$, low slip), Standard (moderate baseline, $P(L_0) \sim \text{Uniform}(0.30, 0.45)$, baseline slip), and Slow (low baseline, $P(L_0) \sim \text{Uniform}(0.15, 0.30)$, higher slip). Profiles were randomized equally into an Experimental Group ($N_e = 30$) and a Control Group ($N_c = 30$).

5.3. Simulation Outcomes and Statistical Rigor

Table 3 details the statistical results, visualized in Figures 2 and 3. The experimental group achieved a mean simulated NLG of 0.779 ± 0.382 versus 0.238 ± 0.688 for the control group ($\Delta\text{NLG} = +0.541$, $p < 0.001$). The relative advantage of adaptive routing varied by archetype, with the Fast archetype exhibiting the largest simulated gain differential ($\Delta\text{NLG} = +0.679$, from 0.010 in control to 0.688 in experimental), followed by Slow ($\Delta\text{NLG} = +0.541$) and Standard ($\Delta\text{NLG} = +0.404$) archetypes; per-archetype control-group variance was high ($\sigma \geq 0.48$ in all three cases), so archetype-level deltas should be read as directional given $n = 10$ profiles per archetype-group cell. A Cohen’s d of 0.973 demonstrates a large effect magnitude under the simulation model.

5.4. Equal-Rate Structural Comparison

To probe whether an experimental-style condition retains an advantage under a structurally independent generative model, we additionally ran a separate, simplified synthetic comparison ($N_e = 30, N_c = 30$) using fixed per-step success-probability formulas ($0.90L + 0.20(1 - L)$ for the experimental-labeled arm vs. $0.85L + 0.15(1 - L)$ for the control-labeled arm) rather than the DC-BKT/bandit mechanism itself. We note explicitly that this comparison does not invoke difficulty routing, the contextual bandit, or

Table 3: Statistical summary of evaluation outcomes ($N = 60$).

Metric	Experimental	Control	Delta
Sample Size	30 profiles	30 profiles	$N = 60$
Pre-Test Mean	$38.2\% \pm 13.7\%$	$36.0\% \pm 12.3\%$	Baseline equalized
Post-Test Mean	$87.2\% \pm 21.5\%$	$52.9\% \pm 39.9\%$	+34.3%
Mean NLG	0.779	0.238	+0.541
Std. Dev. (NLG)	0.382	0.688	Higher var. in control
Welch’s t -test: $t(45.4) = 3.768$, $p = 4.73 \times 10^{-4}$			
Mean diff. 95% CI: [0.252, 0.830]			
Mann-Whitney $U = 709.5$, $p = 9.50 \times 10^{-5}$			
Cohen’s $d = 0.973$, 95% CI: [0.438, 1.508]			

DC-BKT, and its probability formulas are structurally advantaged toward the experimental label by construction; it should therefore not be read as validating the bandit or routing mechanism specifically. It produced $\overline{\text{NLG}}_{\text{exp}} = 0.751 \pm 0.152$ vs. $\overline{\text{NLG}}_{\text{ctrl}} = 0.642 \pm 0.198$ ($\Delta\text{NLG} = +0.109$, Welch’s $t(53.7) = 2.399$, $p = 0.0199$), reproduced exactly from the committed source. We retain it here only for transparency about everything that was originally reported as evidence, and flag it as a methodologically weak comparison rather than an independent robustness check.

5.5. Component-Isolation Ablation: Does Difficulty Routing Help?

The primary result (Section 5.3) and the equal-rate comparison above conflate two mechanisms specific to the experimental condition: a higher active-retrieval checkpoint rate ($p_{\text{learn}} = 0.35$ vs. 0.10) and Thompson-Sampling-based difficulty routing. To isolate the routing mechanism’s own contribution, we re-ran the experimental-arm simulation twice, holding checkpoint rate fixed at 0.35 in both runs: once with bandit-routed difficulty selection (as in the full pipeline), and once with difficulty fixed to medium for every step. This was tested under two independent bandit configurations (the original utility weights $w_a \in \{0.8, 1.0, 1.2\}$, and a milder $w_a \in \{0.9, 1.0, 1.1\}$ configuration, to rule out a single unlucky weight choice), and cross-checked by removing the bandit’s ϵ -greedy random-exploration term entirely.

In every configuration tested, bandit-routed difficulty selection underperformed the fixed-medium baseline: mean NLG 0.878 (bandit, default weights) vs. 0.947 (fixed-medium), a deficit of -0.069 ; the deficit persisted (-0.093) under the milder weight configuration and with ϵ -greedy exploration disabled. We traced the mechanism: within DC-BKT, the same guess/slip parameters used to generate simulated correctness also govern the strength of the Bayesian mastery update, so easier items yield weaker positive evidence per correct answer while harder items are more failure-prone; under the specific guess/slip values used in this paper (Equations 1–3), medium difficulty is close to a locally optimal fixed point, and routing away from it in either direction did not improve measured NLG. We report this as an honest negative finding: under the current reward and utility-weight design, Thompson Sampling difficulty routing is not shown to contribute to the mea-

sured learning-gain advantage; the checkpoint-frequency mechanism (present identically whether routing is used or not) accounts for the primary result. Re-tuning the bandit’s reward shaping and utility weights against this diagnostic is identified as concrete future work rather than treated as resolved.

Collapsing the telemetry-cluster context to a single global state (removing K -Means-derived clustering from the bandit’s state space, all else identical) changed mean NLG from 0.878 to 0.926 (+0.048), and disabling difficulty-conditioning in the BKT update (using fixed $P(G) = 0.20, P(S) = 0.10$ regardless of assigned difficulty) changed it from 0.878 to 0.855 (−0.023). Both deltas are small relative to the routing-vs-fixed-medium effect above and are reported for completeness rather than as strong evidence either way, given they were only evaluated once at $N = 30$ per arm.

5.6. Out-of-Model Robustness: Logistic Regression on Step-Level Data

As an independent check that does not rely on the BKT recursive update at all, we fit a logistic regression directly on the real step-level response records generated by the simulation (mastery level and group membership as predictors of per-step correctness, $n = 360$ observations: 6 steps $\times$ 30 profiles per arm $\times$ 2 arms). Controlling for current mastery (coefficient 2.966), the experimental-group indicator carried a negative coefficient (−0.165), corroborating the ablation finding above through an entirely different statistical method: at matched mastery levels, the bandit-routed experimental arm was not more likely to answer correctly than the fixed-medium arm, consistent with routing toward harder items.

5.7. Multi-Parameter Sensitivity

We perturbed the checkpoint-rate parameter and guess/slip baselines across $2^3 = 8$ combinations (± 0.05 on $P(G)$, $P(S)$, and the checkpoint rate itself), each evaluated once at $N = 60$. Only 1 of 8 configurations retained a statistically significant experimental advantage ($p < 0.05$); the remaining 7 exhibited ceiling saturation, with both arms reaching $NLG \approx 1.0$ within the 6-step horizon and no measurable residual difference. This indicates the primary result, as parameterized in this paper, is not robust to small perturbations of the checkpoint-rate parameter, and should be read as a demonstration under one specific, hand-set parameter regime rather than a robust effect across nearby regimes. A longer evaluation horizon or explicit ceiling-avoidance in the simulator is needed before robustness claims can be made with confidence.

6. Discussion, Practical Implications, and Limitations

6.1. Pedagogical Foundations

CogniLoop’s design is grounded in three cognitive learning principles. Csikszentmihalyi’s Flow Channel motivates Thompson Sampling, which is designed to balance challenge and skill by sampling from updated Beta distributions; we note this is a theoretical design motivation rather than an empirically confirmed effect in this paper, since Section 5.5’s component-isolation ablation did not find a measured NLG benefit from difficulty routing under the tested configurations. Vygotsky’s Zone of Proximal Development (ZPD)

motivates active checkpoints that ensure instruction operates within the learner’s developmental threshold (Bloom, 1984); unlike the routing mechanism above, this principle corresponds directly to the checkpoint-frequency effect that Section 5.5 identifies as the empirically supported driver of the measured gain. Active Retrieval Practice – formative in-video checkpoints enforcing active memory consolidation and blunting the Ebbinghaus decay curve (Karpicke and Roediger, 2008) – is the mechanism most directly supported by our simulation results.

6.2. Methodological Limitations

We explicitly acknowledge the following research boundaries. Simulation vs. Human Efficacy: Phase 1 evaluation parameterizes student behavior within the specified simulation learning environment; while mathematically rigorous, synthetic agents do not capture human fatigue, emotional fluctuations, or external distractions. DC-BKT Hyperparameter Estimation: guess $P(G_a)$ and slip $P(S_a)$ parameters are explicitly hand-set simulation hyperparameters rather than population-fitted estimates from historical learner logs. Domain Scope: the current evaluation focused on modular Computer Science subjects (OS, DS, DBMS, CN); generalization to subjective humanities curricula requires domain-specific prompt engineering. Evaluation Horizon: simulations spanned 6 sequential sessions per agent; longitudinal knowledge retention over multi-month horizons remains an objective for live pilot trials.

Scope and Fragility of Reported Statistics: every statistic in this paper was independently re-executed from committed, seeded simulation source code and reproduces exactly, including the negative and fragility findings in Sections 5.4–5.6. We explicitly flag that the top-line adaptive-vs-static comparison (Section 5.3) does not imply that every named subsystem contributes positively: the component-isolation ablation (Section 5.5) shows Thompson Sampling difficulty routing did not improve measured NLG relative to a fixed-medium baseline under the tested utility-weight configurations, and the sensitivity sweep (Section 5.6) shows the primary effect collapses to ceiling saturation under small perturbations of the checkpoint-rate parameter in 7 of 8 tested regimes. Retrieval-grounding hallucination measurement (which requires real LLM-generated items and human-labeled gold-standard judgments) and validation-threshold F_1 tuning were not evaluated in this revision, as they cannot be responsibly simulated, and remain open empirical work requiring live data collection.

6.3. Enterprise Scalability Roadmap

As an architectural design roadmap for future enterprise deployment (10^7 – 10^8 users): migrating POSIX file-locked JSON repositories to a distributed PostgreSQL cluster with connection pooling; replacing in-process ChromaDB instances with managed vector database clusters; offloading transcript retrieval and vector embedding to asynchronous task workers; and caching prompt embeddings to reduce response times for identical concept requests while preserving API quotas.

7. Conclusion and Future Work

We presented CogniLoop, a closed-loop adaptive learning platform that converts passive open-domain lecture videos into individualized mastery curricula, combining RAG transcript grounding, Difficulty-Conditioned Bayesian Knowledge Tracing, Thompson Sampling contextual bandits, and interaction telemetry clustering into a single architecture. Empirical simulation across $N = 60$ profiles demonstrated a statistically significant improvement in Simulated Normalized Learning Gain under the evaluated environment ($\Delta\text{NLG} = +0.541$, $p < 0.001$, $d = 0.973$). A component-isolation analysis attributes this measured gain specifically to active-retrieval checkpoint frequency; Thompson Sampling difficulty routing, as currently parameterized, did not show a positive contribution in isolation, and the primary effect was not robust to small perturbations of the checkpoint-rate parameter in a multi-parameter sensitivity sweep. We report these mixed findings deliberately: the architecture is a complete, working system that directly solves the content-authoring bottleneck limiting traditional ITS to hand-curated domains, and the checkpoint-based learning gain is real and reproducible, but the difficulty-routing and DC-BKT components require further tuning and evaluation before their individual contributions to learning gain can be claimed.

Future work includes re-tuning the bandit’s reward shaping and utility weights against the component-isolation diagnostic introduced here, extending the sensitivity analysis to a longer evaluation horizon to rule out ceiling-saturation artifacts, conducting Phase 2 IRB-approved live human participant studies in university computer science classrooms, integrating interactive browser-based coding environments (Pyodide), and expanding multi-modal video segmentation.

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Competing Interests. The author has no competing interests to declare.

Author Contributions. K. Aggarwal conceived the research problem, designed the system architecture and evaluation methodology, directed all implementation and analysis (including AI-assisted implementation as declared below), verified all reported results against source code, and approved the final manuscript.

Data Availability. All simulation source code, including the scripts used to generate every statistic reported in this paper, is available at <https://github.com/OmegaKavya/CogniLoop>. The simulation uses synthetically generated learner profiles (random seed 42); no human-subject data was collected or used.

Ethics Statement. This study did not involve human participants, human data, or animal subjects. All learner profiles evaluated in this Phase 1 work are synthetic, algorithmically generated agents; no IRB or ethics board review was required for the work reported in this paper. Phase 2 (Section 1.3) will involve human participants and will undergo IRB approval prior to any data collection.

Declaration of Generative AI Software Tools in the Writing Process

During the preparation of this work, the author(s) used Claude (Anthropic) in Sections 1–7 (drafting of the manuscript text), in Sections 3 and 4 (implementation of the CogniLoop software system described therein, including the BKT engine, contextual bandit policy, and RAG pipeline), and in Sections 2, 5, and 6 (independent re-execution of the simulation source code and re-verification of citations, reported statistics, ablation results, and figures against that source code) in order to draft the manuscript, implement the described system, and identify and correct citation and figure errors present in an earlier draft. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

The author directed the research problem, system design decisions, and scope throughout, and confirms that the statistics reported in Section 5 were independently reproduced from the cited, committed source files. AI tools were not used to fabricate or embellish results; where an earlier AI-assisted draft was found on verification to contain unsupported or incorrect figures, those figures were corrected or removed rather than retained, as documented in Section 6.2.

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